E-Commerce Customer Value Analysis

1. Executive Summary

This project analyzes customer purchasing behavior using the RFM model. SQL was used for data extraction, Python for cleaning and feature engineering, and Power BI for visualization. The objective is to identify valuable customers, reduce churn, and improve business decisions.

2. Business Problem

The company wanted to identify valuable customers, reduce churn, improve retention, and maximize revenue using customer segmentation.

3. Objectives

• Segment customers using RFM
• Identify high-value and lost customers
• Build an interactive Power BI dashboard
• Generate actionable business recommendations

4. Dataset

Brazilian Olist E-Commerce dataset containing customers, orders, payments, products, reviews and sellers.

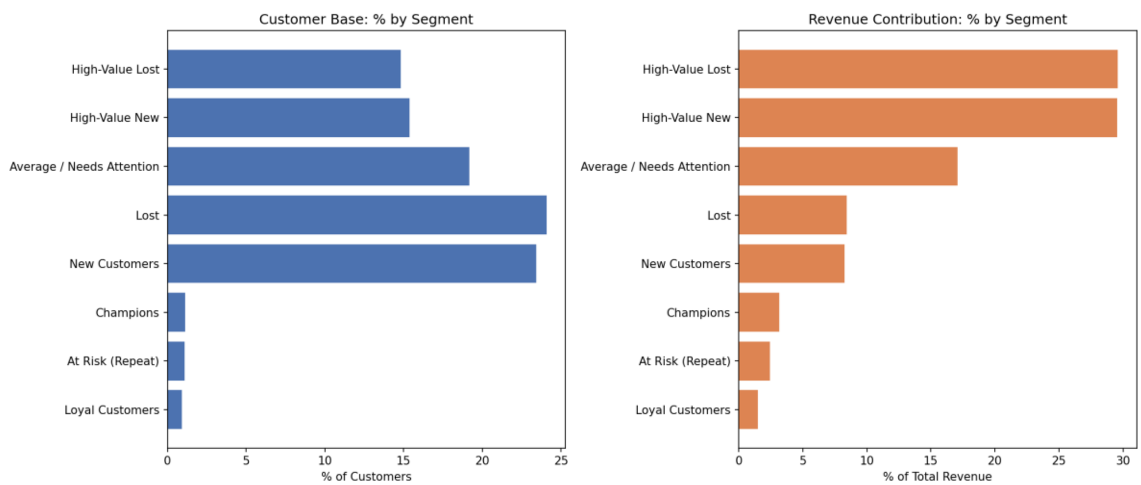
5. Project Workflow

1. SQL Data Extraction
2. Data Cleaning using Pandas
3. RFM Score Calculation
4. Customer Segmentation
5. Dashboard Development in Power BI
6. Business Insights & Recommendations

6. Power BI Dashboard

The dashboard presents KPIs, customer segmentation, revenue analysis, yearly revenue trends, review scores, and customer geographical distribution.

7. Customer Segment Analysis



8. Key Insights

- Lost Customers (24.1%) are the largest customer segment but contribute only 8.4% revenue.
- High-Value New and High-Value Lost customers together generate nearly 60% of total revenue.
- Average customers contribute 17.1% revenue and represent an opportunity for upselling.

- Champion customers are few but extremely valuable for long-term growth.
- New customers indicate strong acquisition performance but require retention efforts.

9. Business Recommendations

- Introduce personalized win-back campaigns for lost customers.
- Launch VIP rewards for high-value customers.
- Implement welcome journeys for new customers.
- Use cross-selling and upselling for average customers.
- Reward champions with loyalty and referral programs.
- Monitor customer health using RFM scores monthly.
- Create targeted email campaigns instead of mass marketing.
- Track customer lifetime value for strategic planning.

10. Challenges & Solutions

Handled missing values, duplicate records, date conversions, customer-level aggregation, and meaningful RFM segmentation.

11. Skills Demonstrated

SQL | Python | Pandas | Data Cleaning | Feature Engineering | RFM Analysis | Power BI | DAX | Data Visualization | Business Analytics

12. Conclusion

Developed an end-to-end customer analytics solution using SQL, Python, and Power BI. Built an interactive dashboard for 96K customers and 99K orders, performed RFM segmentation, and generated business recommendations that support customer retention and revenue growth.